Multi-Turn Speech Dialogue

A curated paper list for Multi-Turn Speech Dialogue.

Survey

1. "Turn-taking in Conversational Systems and Human-Robot Interaction: A Review". *Gabriel Skantze* Computer Speech & Language 2021. Paper
2. "A Review of Dialogue Systems: From Trained Monkeys to Stochastic Parrots". *Atharv Singh Patlan et al.* arXiv 2023. Paper
3. "WavChat: A Survey of Spoken Dialogue Models". *Shengpeng Ji et al.* arXiv 2024. Paper
4. "From Turn-Taking to Synchronous Dialogue: A Survey of Full-Duplex Spoken Language Models". *Yuxuan Chen and Haoyuan Yu* arXiv 2025. Paper
5. "A Survey of Recent Advances on Turn-taking Modeling in Spoken Dialogue Systems". *Galo Castillo-López et al.* IWSDS 2025. Paper
6. "Turn-Taking Modelling in Conversational Systems: A Review of Recent Advances". *Rutherford Agbeshi Patamia et al.* Technologies 2025. Paper
7. "A Survey of Full-Duplex Spoken Dialogue Systems: Architectural Hierarchy, Interaction Ontology, and Decision State Machine". *Jingyu Lu et al.* arXiv 2026. Paper

General Surveys

1. "A Survey of Available Corpora For Building Data-Driven Dialogue Systems: The Journal Version". *Iulian V. Serban et al.* Dialogue & Discourse 2018. Paper
2. "Neural Approaches to Conversational AI". *Jianfeng Gao et al.* Foundations and Trends in Information Retrieval 2019. Paper
3. "Recent Neural Methods on Dialogue State Tracking for Task-Oriented Dialogue Systems: A Survey". *Vevake Balaraman et al.* SIGDIAL 2021. Paper
4. "Who Says What to Whom: A Survey of Multi-Party Conversations". *Jia-Chen Gu et al.* IJCAI 2022. Paper
5. "Transformers in Speech Processing: A Survey". *Siddique Latif et al.* arXiv 2023. Paper
6. "A Survey on Recent Advances in LLM-Based Multi-turn Dialogue Systems". *Zihao Yi et al.* arXiv 2024. Paper
7. "Preference Tuning with Human Feedback on Language, Speech, and Vision Tasks: A Survey". *Genta Indra Winata et al.* arXiv 2024. Paper
8. "A Survey on Speech Large Language Models". *Jing Peng et al.* arXiv 2024. Paper
9. "On The Landscape of Spoken Language Models: A Comprehensive Survey". *Siddhant Arora et al.* arXiv 2025. Paper
10. "Recent Advances in Speech Language Models: A Survey". *Wenqian Cui et al.* ACL 2025. Paper
11. "Aligning Multimodal LLM with Human Preference: A Survey". *Tao Yu et al.* arXiv 2025. Paper

Datasets

1. "Japanese Dialogue Corpus of Information Navigation and Attentive Listening Annotated with Extended ISO-24617-2 Dialogue Act Tags". *Koichiro Yoshino et al.*. LREC 2018. Paper
2. "Taskmaster-1: Toward a Realistic and Diverse Dialog Dataset". *Bill Byrne et al.*. EMNLP-IJCNLP 2019. Paper
3. "Alexa in the Wild: Collecting Unconstrained Conversations with a Modern Voice Assistant in a Public Environment". *Benjamin Benk et al.*. LREC 2020. Paper
4. "HarperValleyBank: A Domain-Specific Spoken Dialog Corpus". *Mike Wu et al.*. arXiv 2020. Paper
5. "The MSP-Conversation Corpus". *Luz Martinez-Lucas et al.*. Interspeech 2020. Paper
6. "ASCEND: A Spontaneous Chinese-English Dataset for Code-switching in Multi-turn Conversation". *Holy Lovenia et al.*. arXiv 2021. Paper
7. "DailyTalk: Spoken Dialogue Dataset for Conversational Text-to-Speech". *Keon Lee et al.* arXiv 2022. Paper
8. "DOROTHIE: Spoken Dialogue for Handling Unexpected Situations in Interactive Autonomous Driving Agents". *Ziqiao Ma et al.* EMNLP Findings 2022. Paper
9. "Collection and Analysis of Travel Agency Task Dialogues with Age-Diverse Speakers". *Michimasa Inaba et al.*. LREC 2022. Paper
10. "Design and Evaluation of the Corpus of Everyday Japanese Conversation". *Hanae Koiso et al.*. LREC 2022. Paper
11. "SpokenWOZ: A Large-Scale Speech-Text Benchmark for Spoken Task-Oriented Dialogue Agents". *Shuzheng Si et al.* arXiv 2023. Paper
12. "CALLS: Japanese Empathetic Dialogue Speech Corpus of Complaint Handling and Attentive Listening in Customer Center". *Yuki Saito et al.* Interspeech 2023. Paper
13. "Advancing Large Language Models to Capture Varied Speaking Styles and Respond Properly in Spoken Conversations (StyleTalk)". *Guan-Ting Lin et al.* ACL 2024. Paper
14. "Generative Expressive Conversational Speech Synthesis (NCSSD)". *Rui Liu et al.* ACM MM 2024. Paper
15. "J-CHAT: Japanese Large-scale Spoken Dialogue Corpus for Spoken Dialogue Language Modeling". *Wataru Nakata et al.* arXiv 2024. Paper
16. "DeepDialogue: A Multi-Turn Emotionally-Rich Spoken Dialogue Dataset". *Alkis Koudounas et al.* arXiv 2025. Paper
17. "UltraVoice: Scaling Fine-Grained Style-Controlled Speech Conversations for Spoken Dialogue Models". *Wenming Tu et al.* arXiv 2025. Paper
18. "Toward Conversational Hungarian Speech Recognition: Introducing the BEA-Large and BEA-Dialogue Datasets". *Mate Gedeon et al.* arXiv 2025. Paper
19. "InteractSpeech: A Speech Dialogue Interaction Corpus for Spoken Dialogue Model". *Yifu Chen et al.* EMNLP Findings 2025. Paper
20. "Behavior-SD: Behaviorally Aware Spoken Dialogue Generation with Large Language Models". *Sehun Lee et al.* NAACL 2025. Paper

21. "RealTalk-CN: A Realistic Chinese Speech-Text Dialogue Benchmark With Cross-Modal Interaction Analysis". *Enzhi Wang et al.* arXiv 2025. Paper
22. "DialogueAgents: A Hybrid Agent-Based Speech Synthesis Framework for Multi-Party Dialogue". *Zheyuan Zhang et al.* arXiv 2025. Paper
23. "Open-Source Full-Duplex Conversational Datasets for Natural and Interactive Speech Synthesis". *Rui Liu et al.* arXiv 2025. Paper
24. "Exploring Emotional Nuances in Spoken Dialogue: Dataset Construction and Prediction of Emotional Dialogue Breakdown". *Hyuga Nakaguro et al.* IWSDS 2026. Paper
25. "DuplexChat: Constructing Speaker-Separated Full-Duplex Dialogue Speech at Scale for Spoken Dialogue Language Modeling". *Wataru Nakata et al.* arXiv 2026. Paper
26. "Dial HEALTHDIAL for Advice: A Multilingual and Multi-Parallel Spoken Dialogue Dataset for Knowledge-Grounded Information Seeking". *Songbo Hu et al.* ACL Findings 2026. Paper

Evaluation & Benchmarks

Evaluation Focus: Turn-taking & Interruption

1. "Investigating Speech Features for Continuous Turn-Taking Prediction Using LSTMs". *Matthew Roddy et al..* Interspeech 2018. Paper
2. "Turn-Taking Predictions Across Languages and Genres Using an LSTM Recurrent Neural Network". *Nigel G. Ward et al..* SLT 2018. Paper
3. "How Much Does Prosody Help Turn-taking? Investigations using Voice Activity Projection Models". *Erik Ekstedt et al.* SIGDIAL 2022. Paper
4. "Talking Turns: Benchmarking Audio Foundation Models on Turn-Taking Dynamics". *Siddhant Arora et al.* ICLR 2025. Paper
5. "Investigating the Impact of Incremental Processing and Voice Activity Projection on Spoken Dialogue Systems". *Yuya Chiba et al.* COLING 2025. Paper
6. "Synchronization and Turn-Taking in Full-Duplex Speech Dialogue Models". *Pablo Riera et al.* arXiv 2026. Paper
7. "Still Between Us? Evaluating and Improving Voice Assistant Robustness to Third-Party Interruptions". *Dongwook Lee et al.* ACL 2026. Paper
8. "Rethinking Binary Evaluation of Turn-Taking under Inherent Ambiguity". *Yunosuke Kubo et al.* SIGDIAL 2026. Paper

Evaluation Focus: Full-duplex Interaction

1. "MMedFD: A Real-world Healthcare Benchmark for Multi-turn Full-Duplex Automatic Speech Recognition". *Hongzhao Chen et al.* arXiv 2025. Paper
2. "Full-Duplex-Bench: A Benchmark to Evaluate Full-duplex Spoken Dialogue Models on Turn-taking Capabilities". *Guan-Ting Lin et al.* arXiv 2025. Paper
3. "FD-Bench: A Full-Duplex Benchmarking Pipeline Designed for Full Duplex Spoken Dialogue Systems". *Yizhou Peng et al.* Interspeech 2025. Paper

4. "The ICASSP 2026 HumDial Challenge: Benchmarking Human-like Spoken Dialogue Systems in the LLM Era". *Zhixian Zhao et al.* ICASSP / arXiv 2026. Paper
5. "τ-Voice: Benchmarking Full-Duplex Voice Agents on Real-World Domains". *Soham Ray et al.* ICML 2026. Paper
6. "Full-Duplex-Bench-v3: Benchmarking Tool Use for Full-Duplex Voice Agents Under Real-World Disfluency". *Guan-Ting Lin et al.* arXiv 2026. Paper
7. "Full-Duplex Interaction in Spoken Dialogue Systems: A Comprehensive Study from the ICASSP 2026 HumDial Challenge". *Chengyou Wang et al.* arXiv 2026. Paper
8. "Full-Duplex-Bench-v2: A Multi-Turn Evaluation Framework for Duplex Dialogue Systems with an Automated Examiner". *Guan-Ting Lin et al.* ACL 2026. Paper
9. "MTR-DuplexBench: Towards a Comprehensive Evaluation of Multi-Round Conversations for Full-Duplex Speech Language Models". *Zhang He et al.* ACL Findings 2026. Paper
10. "M3-DuplexBench: A Multi-Turn, Multilingual, Multidomain Benchmark for Full-Duplex Spoken Dialogue Models". *Ryo Fukuda et al.* arXiv 2026. Paper

Evaluation Focus: Multi-turn Dialogue Capabilities

1. "SD-Eval: A Benchmark Dataset for Spoken Dialogue Understanding Beyond Words". *Junyi Ao et al.* NeurIPS 2024 (Datasets & Benchmarks). Paper
2. "Contextual Interactive Evaluation of TTS Models in Dialogue Systems". *Siyang Wang et al.* Interspeech 2024. Paper
3. "Benchmarking Open-ended Audio Dialogue Understanding for Large Audio-Language Models". *Kuofeng Gao et al.* arXiv 2024. Paper
4. "URO-Bench: Towards Comprehensive Evaluation for End-to-End Spoken Dialogue Models". *Ruiqi Yan et al.* EMNLP Findings 2025. Paper
5. "WavReward: Spoken Dialogue Models With Generalist Reward Evaluators". *Shengpeng Ji et al.* arXiv 2025. Paper
6. "MAD: A Benchmark for Multi-Turn Audio Dialogue Fact-Checking". *Chaewan Chun et al.* arXiv 2025. Paper
7. "VoxRole: A Comprehensive Benchmark for Evaluating Speech-Based Role-Playing Agents". *Weihao Wu et al.* arXiv 2025. Paper
8. "Evaluating Bias in Spoken Dialogue LLMs for Real-World Decisions and Recommendations (FairDialogue)". *Yihao Wu et al.* arXiv 2025. Paper
9. "MULTI-Bench: A Multi-turn Interactive Benchmark for Assessing Emotional Intelligence Ability of Spoken Dialogue Models". *Yayue Deng et al.* arXiv 2025. Paper
10. "Audio MultiChallenge: A Multi-Turn Evaluation of Spoken Dialogue Systems on Natural Human Interaction". *Advait Gosai et al.* arXiv 2025. Paper
11. "VoxDialogue: Can Spoken Dialogue Systems Understand Information Beyond Words?". *Xize Cheng et al.* ICLR 2025. Paper
12. "EchoMind: An Interrelated Multi-level Benchmark for Evaluating Empathetic Speech Language Models". *Li Zhou et al* arXiv 2025. Paper
13. "SOVA-Bench: Benchmarking the Speech Conversation Ability for LLM-based Voice Assistant". *Yixuan Hou et al.* Interspeech 2025. Paper

14. "MTalk-Bench: Evaluating Speech-to-Speech Models in Multi-Turn Dialogues via Arena-style and Rubrics Protocols". *Yuhao Du et al.* arXiv 2025. Paper
15. "VoiceAgentBench: Are Voice Assistants Ready for Agentic Tasks?". *Dhruv Jain et al.* arXiv 2025. Paper
16. "C3: A Bilingual Benchmark for Spoken Dialogue Models Exploring Challenges in Complex Conversations". *Chengqian Ma et al.* EMNLP 2025. Paper
17. "HumDial-EIBench: A Human-Recorded Multi-Turn Emotional Intelligence Benchmark for Audio Language Models". *Shuiyuan Wang et al* arXiv 2026. Paper
18. "VoiceBench: Benchmarking LLM-Based Voice Assistants". *Yiming Chen et al.* TACL 2026. Paper
19. "The Context Trap: Why End-to-End Audio Language Models Fail Multi-turn Dialogues". *Zhi Rui Tam et al.* IWSDS 2026. Paper
20. "SDiaReward: Modeling and Benchmarking Spoken Dialogue Rewards with Modality and Colloquialness". *Jingyu Lu et al.* ACL 2026. Paper
21. "SpeakerSleuth: Can Large Audio-Language Models Judge Speaker Consistency across Multi-turn Dialogues?". *Jonggeun Lee et al.* ACL 2026. Paper
22. "Style Amnesia: Investigating Speaking Style Degradation and Mitigation in Multi-Turn Spoken Language Models". *Yu-Xiang Lin et al.* ACL Findings 2026. Paper

Models

Interaction Mode: Turn-taking

1. "Improving End-of-Turn Detection in Spoken Dialogues by Detecting Speaker Intentions as a Secondary Task". *Zakaria Aldeneh et al..* ICASSP 2018. Paper
2. "Neural Dialogue Context Online End-of-Turn Detection". *Ryo Masumura et al..* SIGDIAL 2018. Paper
3. "Prediction of Turn-Taking Using Multitask Learning with Prediction of Backchannels and Fillers". *Kohei Hara et al..* Interspeech 2018. Paper
4. "Turn-Taking Prediction Based on Detection of Transition Relevance Place". *Kohei Hara et al..* Interspeech 2019. Paper
5. "TurnGPT: A Transformer-Based Language Model for Predicting Turn-Taking in Spoken Dialog". *Erik Ekstedt and Gabriel Skantze.* EMNLP Findings 2020. Paper
6. "Projection of Turn Completion in Incremental Spoken Dialogue Systems". *Erik Ekstedt and Gabriel Skantze.* SIGDIAL 2021. Paper
7. "Timing Generating Networks: Neural Network Based Precise Turn-Taking Timing Prediction in Multiparty Conversation". *Shinya Fujie et al..* Interspeech 2021. Paper
8. "Using Transition Duration to Improve Turn-taking in Conversational Agents". *Charles Threlkeld et al.* SIGDIAL 2022. Paper
9. "When can I Speak? Predicting Initiation Points for Spoken Dialogue Agents". *Siyan Li et al.* SIGDIAL 2022. Paper

10. "Voice Activity Projection: Self-supervised Learning of Turn-taking Events". *Erik Ekstedt and Gabriel Skantze* Interspeech 2022. Paper
11. "Gated Multimodal Fusion with Contrastive Learning for Turn-Taking Prediction in Human-Robot Dialogue". *Jiudong Yang et al..* ICASSP 2022. Paper
12. "Response Timing Estimation for Spoken Dialog System Using Dialog Act Estimation". *Jin Sakuma et al..* Interspeech 2022. Paper
13. "Turn-Taking Prediction for Natural Conversational Speech". *Shuo-Yiin Chang et al..* Interspeech 2022. Paper
14. "Generative Spoken Dialogue Language Modeling". *Tu Anh Nguyen et al.* arXiv 2022 / TACL 2023. Paper
15. "SpeechGPT: Empowering Large Language Models with Intrinsic Cross-Modal Conversational Abilities". *Dong Zhang et al.* EMNLP Findings 2023. Paper
16. "Let's Go Real Talk: Spoken Dialogue Model for Face-to-Face Conversation". *Se Park et al.* ACL 2024. Paper
17. "Qwen2-Audio Technical Report". *Yunfei Chu et al.* arXiv 2024. Paper
18. "Style-Talker: Finetuning Audio Language Model and Style-Based Text-to-Speech Model for Fast Spoken Dialogue Generation". *Yinghao Aaron Li et al.* arXiv 2024. Paper
19. "EMOVA: Empowering Language Models to See, Hear and Speak with Vivid Emotions". *Kai Chen et al.* arXiv 2024. Paper
20. "Internalizing ASR with Implicit Chain of Thought for Efficient Speech-to-Speech Conversational LLM". *Robin Shinghei Yuen et al.* NeurIPS 2024. Paper
21. "Building a Taiwanese Mandarin Spoken Language Model: A First Attempt". *Chih-Kai Yang et al.* arXiv 2024. Paper
22. "LLaMA-Omni: Seamless Speech Interaction with Large Language Models". *Qingkai Fang et al.* arXiv 2024. Paper
23. "Mini-Omni: Language Models Can Hear, Talk While Thinking in Streaming". *Zhifei Xie et al.* arXiv 2024. Paper
24. "GLM-4-Voice: Towards Intelligent and Human-Like End-to-End Spoken Chatbot". *Aohan Zeng et al.* arXiv 2024. Paper
25. "Triadic Multi-party Voice Activity Projection for Turn-taking in Spoken Dialogue Systems". *Mikey Elmers et al.* Interspeech 2025. Paper
26. "OpenOmni: Advancing Open-Source Omnimodal Large Language Models with Progressive Multimodal Alignment and Real-time Emotional Speech Synthesis". *Run Luo et al.* arXiv 2025. Paper
27. "GOAT-SLM: A Spoken Language Model with Paralinguistic and Speaker Characteristic Awareness". *Hongjie Chen et al.* arXiv 2025. Paper
28. "OpenS2S: Advancing Fully Open-Source End-to-End Empathetic Large Speech Language Model". *Chen Wang et al.* arXiv 2025. Paper
29. "Step-Audio 2 Technical Report". *StepFun Team et al.* arXiv 2025. Paper
30. "InteractiveOmni: A Unified Omni-modal Model for Audio-Visual Multi-turn Dialogue". *Wenwen Tong et al.* arXiv 2025. Paper

31. "Step-Audio: Unified Understanding and Generation in Intelligent Speech Interaction". *Ailin Huang et al.* arXiv 2025. Paper
32. "Baichuan-Audio: A Unified Framework for End-to-End Speech Interaction". *Tianpeng Li et al.* arXiv 2025. Paper
33. "Kimi-Audio Technical Report". *KimiTeam et al.* arXiv 2025. Paper
34. "Qwen3-Omni Technical Report". *Jin Xu et al.* arXiv 2025. Paper
35. "PACHAT: Persona-Aware Speech Assistant for Multi-party Dialogue". *Dongjie Fu et al.* EMNLP 2025. Paper
36. "VITA-Audio: Fast Interleaved Cross-Modal Token Generation for Efficient Large Speech-Language Model". *Zuwei Long et al.* arXiv 2025. Paper
37. "AV-Dialog: Spoken Dialogue Models with Audio-Visual Input". *Tuochao Chen et al.* arXiv 2025. Paper
38. "Qwen2.5-Omni Technical Report". *Qwen Team et al.* arXiv 2025. Paper
39. "LLaMA-Omni 2: LLM-based Real-time Spoken Chatbot with Autoregressive Streaming Speech Synthesis". *Qingkai Fang et al.* ACL 2025. Paper
40. "VoxMind: An End-to-End Agentic Spoken Dialogue System". *Tianle Liang et al.* ACL 2026. Paper
41. "ZipVoice-Dialog: Non-Autoregressive Spoken Dialogue Generation with Flow Matching". *Han Zhu et al.* ACL Findings 2026. Paper

Interaction Mode: Half-duplex / Controlled Barge-in

1. "An Incremental Turn-Taking Model for Task-Oriented Dialog Systems". *Andrei C. Coman et al..* Interspeech 2019. Paper
2. "Oh, Jeez! or Uh-Huh? A Listener-Aware Backchannel Predictor on ASR Transcriptions". *Daniel Ortega et al..* ICASSP 2020. Paper
3. "BPM_MT: Enhanced Backchannel Prediction Model Using Multi-Task Learning". *Jin Yea Jang et al..* EMNLP 2021. Paper
4. "Device Directedness with Contextual Cues for Spoken Dialog Systems". *Dhanush Bekal et al.* arXiv 2022. Paper
5. "Contextual Acoustic Barge-In Classification for Spoken Dialog Systems". *Dhanush Bekal et al..* Interspeech 2022. Paper
6. "Freeze-Omni: A Smart and Low Latency Speech-to-Speech Dialogue Model with Frozen LLM". *Xiong Wang et al.* arXiv 2024. Paper
7. "Mini-Omni2: Towards Open-source GPT-4o with Vision, Speech and Duplex Capabilities". *Zhifei Xie et al.* arXiv 2024. Paper
8. "VITA: Towards Open-Source Interactive Omni Multimodal LLM". *Chaoyou Fu et al.* arXiv 2024. Paper
9. "SHANKS: Simultaneous Hearing and Thinking for Spoken Language Models". *Cheng-Han Chiang et al.* arXiv 2025. Paper

Interaction Mode: Full-duplex

1. "Duplex Conversation in Outbound Agent System". *Chunxiang Jin et al.* Interspeech 2021. Paper

2. "Duplex Conversation: Towards Human-like Interaction in Spoken Dialogue Systems". *Ting-En Lin et al.* arXiv 2022. Paper
3. "Beyond Turn-Based Interfaces: Synchronous LLMs as Full-Duplex Dialogue Agents". *Bandhav Veluri et al.* EMNLP 2024. Paper
4. "Moshi: A Speech-Text Foundation Model for Real-Time Dialogue". *Alexandre Defossez et al.* arXiv 2024. Paper
5. "SALMONN-omni: A Codec-free LLM for Full-duplex Speech Understanding and Generation". *Wenyi Yu et al.* arXiv 2024. Paper
6. "A Full-duplex Speech Dialogue Scheme Based On Large Language Models". *Peng Wang et al.* NeurIPS 2024. Paper
7. "MinMo: A Multimodal Large Language Model for Seamless Voice Interaction". *FunAudioLLM Team et al.* arXiv 2025. Paper
8. "SALM-Duplex: Efficient and Direct Duplex Modeling for Speech-to-Speech Language Model". *Ke Hu et al.* arXiv 2025. Paper
9. "OmniFlatten: An End-to-end GPT Model for Seamless Voice Conversation". *Qinglin Zhang et al.* ACL 2025. Paper
10. "Towards a Japanese Full-duplex Spoken Dialogue System". *Atsumoto Ohashi et al.* Interspeech 2025. Paper
11. "FireRedChat: A Pluggable, Full-Duplex Voice Interaction System with Cascaded and Semi-Cascaded Implementations". *Junjie Chen et al.* arXiv 2025. Paper
12. "Incorporating Dialogue State Tracking into Japanese Full-duplex Task-oriented Spoken Dialogue Model". *Yuya Chiba et al.* IJCNLP-AACL 2025. Paper
13. "Chronological Thinking in Full-Duplex Spoken Dialogue Language Models". *Donghang Wu et al.* arXiv 2025. Paper
14. "LLM-Enhanced Dialogue Management for Full-Duplex Spoken Dialogue Systems". *Hao Zhang et al.* arXiv 2025. Paper
15. "Language Model Can Listen While Speaking". *Ziyang Ma et al.* AAAI 2025. Paper
16. "FlashLabs Chroma 1.0: A Real-Time End-to-End Spoken Dialogue Model with Personalized Voice Cloning". *Tanyu Chen et al.* arXiv 2026. Paper
17. "PersonaPlex: Voice and Role Control for Full Duplex Conversational Speech Models". *Rajarshi Roy et al.* arXiv 2026. Paper
18. "MiniCPM-o 4.5: Towards Real-Time Full-Duplex Omni-Modal Interaction". *Junbo Cui et al.* arXiv 2026. Paper
19. "DuplexSLA: A Full-Duplex Spoken Language Model with Synchronized Speech, Language, and Action". *Haoyang Zhang et al.* arXiv 2026. Paper
20. "BayLing-Duplex: Native Full-Duplex Speech Dialogue with a Single Autoregressive LLM". *Qingkai Fang et al.* arXiv 2026. Paper
21. "F-Actor: Controllable Conversational Behavior in Full-Duplex Models". *Maike Züfle et al.* ACL Findings 2026. Paper
22. "Hierarchical Acoustic-Semantic Modeling: Modality Separation and Semantic Coherence for Full-Duplex SLMs". *Zhenyu Liu et al.* ACL 2026. Paper

23. "JoyAI-Talker: Full-Duplex Speech Interactive Large Model Built for Empathetic Voice Agents". *Yinhao Bai et al.* arXiv 2026. Paper
24. "SoulX-Duplug: Plug-and-Play Streaming State Prediction Module for Realtime Full-Duplex Speech Conversation". *Ruiqi Yan et al.* arXiv 2026. Paper

Training Methods

Interaction Mode: Turn-taking

1. "Towards Generalized Models for Task-Oriented Dialogue Modeling on Spoken Conversations". *Ruijie Yan et al.*. arXiv 2022. Paper
2. "Adapting Text-based Dialogue State Tracker for Spoken Dialogues". *Jaeseok Yoon et al.* DSTC 2023. Paper
3. "SpeechAlign: Aligning Speech Generation to Human Preferences". *Dong Zhang et al.* NeurIPS 2024. Paper
4. "Data-Centric Improvements for Enhancing Multi-Modal Understanding in Spoken Conversation Modeling (ASK-QA)". *Maximillian Chen et al.* ACL Findings 2025. Paper
5. "WavRAG: Audio-Integrated Retrieval Augmented Generation for Spoken Dialogue Models". *Yifu Chen et al.* ACL 2025. Paper
6. "Enhancing Speech-to-Speech Dialogue Modeling with End-to-End Retrieval-Augmented Generation". *Pengchao Feng et al.* EMNLP Findings 2025. Paper
7. "Aligning Spoken Dialogue Models from User Interactions". *Anne Wu et al.* arXiv 2025. Paper
8. "Chain-of-Thought Training for Open E2E Spoken Dialogue Systems". *Siddhant Arora et al.* arXiv 2025. Paper
9. "VoiceTextBlender: Augmenting Large Language Models with Speech Capabilities via Single-Stage Joint Speech-Text Supervised Fine-Tuning". *Yifan Peng et al.* NAACL 2025. Paper
10. "Approaching Dialogue State Tracking via Aligning Speech Encoders and LLMs". *Šimon Sedláček et al.* arXiv 2025. Paper
11. "The Speech-LLM Takes It All: A Truly Fully End-to-End Spoken Dialogue State Tracking Approach". *Nizar El Ghazal et al.* arXiv 2025. Paper
12. "Optimizing Conversational Quality in Spoken Dialogue Systems with Reinforcement Learning from AI Feedback". *Siddhant Arora et al.* arXiv 2026. Paper
13. "WavAlign: Enhancing Intelligence and Expressiveness in Spoken Dialogue Models via Adaptive Hybrid Post-Training". *Yifu Chen et al.* ACL Findings 2026. Paper

Interaction Mode: Half-duplex / Streaming

1. "A Methodology for Turn-Taking Capabilities Enhancement in Spoken Dialogue Systems Using Reinforcement Learning". *Hatim Khouzaimi et al.*. Computer Speech & Language 2018. Paper
2. "Stream RAG: Instant and Accurate Spoken Dialogue Systems with Streaming Tool Usage". *Siddhant Arora et al.* arXiv 2025. Paper
3. "Dual-Reasoner: Bridging Interleaved Atomicity and Streaming Latency via Thinking-while-Talking". *Yangzhuo Li et al.* ACL Findings 2026. Paper

Interaction Mode: Full-duplex

1. "FLM-Audio: Natural Monologues Improves Native Full-Duplex Chatbots via Dual Training". *Yiqun Yao et al.* arXiv 2025. Paper
2. "Reinforcement Learning Enhanced Full-Duplex Spoken Dialogue Language Models for Conversational Interactions". *Chen Chen et al.* COLM 2025. Paper
3. "Multi-Faceted Interactivity Alignment in Full-Duplex Speech Models". *Atsumoto Ohashi et al.* arXiv 2026. Paper
4. "Reproducing Proficiency-Conditioned Dialogue Features with Full-duplex Spoken Dialogue Models". *Takao Obi et al.* IWSDS 2026. Paper
5. "Effects of Dialogue Corpora Properties on Fine-Tuning a Moshi-Based Spoken Dialogue Model". *Yuto Abe et al.* IWSDS 2026. Paper
6. "Dual-Axis Generative Reward Model Toward Semantic and Turn-taking Robustness in Interactive Spoken Dialogue Models". *Yifu Chen et al.* ACL 2026. Paper
7. "Sommelier: Scalable Open Multi-turn Audio Pre-processing for Full-duplex Speech Language Models". *Kyudan Jung et al.* ACL Industry Track 2026. Paper

Survey Coverage Comparison

Table 1. Coverage comparison of surveys related to Multi-Turn Speech Dialogue, including fine-grained organization aligned with this collection.

General Surveys are listed first, followed by core surveys. ✓ = systematic coverage; △ = partial or supporting coverage; - = not a substantive focus. Interaction Taxonomy denotes an explicit taxonomy of conversational interaction. Fine-grained Organization assesses whether a survey structures its analysis using the evaluation focuses, model interaction modes, and training interaction modes adopted in this collection.

Survey	Research Scope			Content Coverage				Interaction Taxonomy	Fine-grained Organization		
	Multi-turn Speech	Turn-taking & Interruption	Full-duplex	Datasets	Evaluation / Benchmark	Models	Training Methods		Evaluation Focus	Model Interaction Modes	Training Interaction Modes
Serban et al. (2018)	△	-	-	✓	✓	△	△	✓	-	-	-
Gao et al. (2019)	△	-	-	△	△	✓	✓	✓	-	-	-
Balaraman et al. (2021)	△	-	-	✓	✓	✓	△	✓	△	-	-
Gu et al. (2022)	△	△	-	△	△	✓	△	✓	△	△	-
Latif et al. (2023)	△	-	-	△	△	✓	△	✓	-	-	-
Yi et al. (2024)	△	-	-	✓	✓	✓	✓	△	△	-	△
Winata et al. (2024)	-	-	-	✓	✓	△	✓	✓	-	-	△
Peng et al. (2024)	△	-	-	△	✓	✓	✓	✓	△	△	△
Arora et al. (2025)	△	△	△	△	✓	✓	✓	✓	△	△	△
Cui et al. (2025)	△	-	△	△	✓	✓	✓	✓	△	△	△
Yu et al. (2025)	-	-	-	✓	✓	△	✓	✓	-	-	△
Skantze (2021)	✓	✓	△	△	△	✓	-	✓	✓	✓	△
Patlan et al. (2023)	✓	△	-	△	✓	✓	✓	✓	△	△	△
Ji et al. / WavChat (2024)	✓	✓	✓	✓	✓	✓	✓	✓	△	△	△
Chen and Yu (2025)	✓	✓	✓	△	✓	✓	△	✓	✓	✓	△
Castillo-López et al. (2025)	✓	✓	-	✓	△	✓	△	✓	✓	✓	✓
Patamia et al. (2025)	✓	✓	△	✓	✓	✓	△	✓	✓	✓	✓
Lu et al. (2026)	✓	✓	✓	✓	✓	✓	△	✓	✓	✓	△
Current Collection	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

System and Model Comparison

Table 2. Comparison of all papers listed under Models for Multi-Turn Speech Dialogue.

✓ = explicitly reported support; △ = partial, asymmetric, or controller-mediated support; - = not reported, not applicable, or not a primary capability. Method names link to the papers.

Method	Scope	Interaction Capabilities				Modalities		System Architecture			Speech Interface	Acoustic / Speech Model	Release
		Streaming	Barge-in	Concurrent	Backchannel / Overlap	Speech Output	Vision	Cascaded	Semi-casc.	End-to-end			
Interaction Mode: Turn-taking													
EOT + Intent	Component	△	-	-	-	-	-	-	-	-	Acoustic features	BiLSTM acoustic EOT/intent classifier	2018.04
Neural Online EOT	Component	✓	-	-	-	-	-	-	-	-	Acoustic + context	LSTM online EOT predictor (no generator)	2018.07
MTL Turn-taking	Component	△	-	-	✓	-	-	-	-	-	Acoustic features	Multitask LSTM turn/BC/filler predictor	2018.09
TRP Turn-taking	Component	△	-	-	-	-	-	-	-	-	Speech features	LSTM TRP predictor (no generator)	2019.09
TurnGPT	Component	△	-	-	-	-	-	-	-	-	Text tokens	GPT-2 turn-shift LM (text-only)	2020.11
Turn Completion Projection	Component	✓	-	-	-	-	-	-	-	-	Incremental text	Incremental completion predictor (no generator)	2021.07
Timing Generating Networks	Component	△	-	-	-	-	-	-	-	-	Timing features	LSTM timing-generating network	2021.08
Transition Duration	Component	△	-	-	-	-	-	-	-	-	Timing features	Transition-duration predictor (no generator)	2022.09
Initiation Point Predictor	Component	△	-	-	-	-	-	-	-	-	Speech + context	Neural initiation-point predictor (no generator)	2022.09
VAP	Component	✓	-	-	✓	-	-	-	-	-	Waveform	Causal Transformer voice-activity predictor	2022.09
Gated Multimodal Fusion	Component	△	-	-	-	-	✓	-	-	-	Audio-visual features	Gated audio-visual turn predictor	2022.05
Response Timing Estimator	Component	△	-	-	-	-	-	-	-	-	Dialogue acts	DNN response-timing estimator	2022.09
Natural TT Predictor	Component	△	-	-	-	-	-	-	-	-	Conversational speech	Neural turn-taking predictor (no generator)	2022.09
dGSLM	Model	△	-	△	✓	✓	-	-	-	✓	HuBERT units	HuBERT units + dual-stream Transformer + HiFi-GAN	2022.03
SpeechGPT	Model	-	-	-	-	✓	-	-	-	✓	HuBERT units	HuBERT units + LLaMA + Unit HiFi-GAN	2023.05
RealTalk	System	△	-	-	-	✓	-	-	-	✓	Speech features	SSL speech encoder + dialogue Transformer + unit decoder	2024.08
Qwen2-Audio	Model	-	-	-	-	-	-	-	✓	-	Whisper features	Whisper-large-v3 encoder + Qwen-7B	2024.07
Style-Talker	System	△	-	-	-	✓	-	-	✓	-	Audio LM + TTS	Speech encoder + audio LM + style-conditioned TTS	2024.08
EMOVA	Model	-	-	-	-	✓	✓	-	-	✓	Semantic + acoustic	Semantic-acoustic tokenizer + LLM + style decoder	2024.09
IntrinsicVoice	Model	△	-	-	-	✓	-	-	-	✓	HuBERT units	HuBERT units + LLaMA + GroupFormer decoder	2024.09
Taiwanese Mandarin SLM	Model	-	-	-	-	✓	-	-	-	✓	Discrete units	Discrete-unit Transformer LM + unit vocoder	2024.11
LLaMA-Omni	System	△	-	-	-	✓	-	-	-	✓	Whisper + units	Whisper encoder + LLaMA + streaming unit decoder	2024.09
Mini-Omni	System	△	-	-	-	✓	-	-	-	✓	SNAC codec	Whisper encoder + Qwen + SNAC token decoder	2024.08
GLM-4-Voice	System	✓	-	-	-	✓	-	-	-	✓	Speech tokenizer	Speech tokenizer + GLM-4 + flow-matching decoder	2024.12
Triadic VAP	Component	✓	-	-	✓	-	-	-	-	-	Multi-party waveform	Causal Transformer triadic VAP predictor	2025.08
OpenOmni	Model	△	-	-	-	✓	✓	-	-	✓	Speech tokens	Pretrained speech encoder + LLM + lightweight speech decoder	2025.01
GOAT-SLM	Model	-	-	-	-	✓	-	-	-	✓	Speech tokens	Dual semantic/acoustic heads + speech decoder	2025.07
OpenS2S	Model	-	-	-	-	✓	-	-	-	✓	Speech tokens	BLSP-Emo + streaming interleaved speech decoder	2025.07
Step-Audio 2	System	✓	-	-	-	✓	-	-	-	✓	Speech-text tokens	Step-Audio tokenizer + LLM + speech-token decoder	2025.07
InteractiveOmni	Model	△	-	-	-	✓	✓	-	-	✓	Audio-visual tokens	Audio-visual encoders + unified Transformer decoder	2025.10
Step-Audio	System	✓	-	-	-	✓	-	-	-	✓	Speech-text tokens	Step-Audio tokenizer + 130B LLM + speech decoder	2025.02
Baichuan-Audio	System	✓	-	-	-	✓	-	-	-	✓	Multi-codebook	Speech encoder + Baichuan LLM + multi-codebook decoder	2025.02
Kimi-Audio	Model	✓	-	-	-	✓	-	-	-	✓	Continuous + tokens	Audio encoder/tokenizer + Kimi LLM + speech decoder	2025.04
Qwen3-Omni	Model	✓	-	-	-	✓	✓	-	-	✓	Continuous + codec	Qwen Thinker + autoregressive Talker decoder	2025.09
PACHAT	System	△	-	-	✓	✓	-	-	✓	-	Speech frontend + TTS	Speech frontend + persona-aware LLM + expressive TTS	2025.11
VITA-Audio	Model	✓	-	-	-	✓	-	-	-	✓	Interleaved tokens	Audio encoder + LLM + interleaved speech-token decoder	2025.05
AV-Dialog	Model	-	-	-	-	✓	✓	-	-	✓	Audio-visual features	Audio-visual encoders + dialogue Transformer + speech decoder	2025.11
Qwen2.5-Omni	Model	✓	-	-	-	✓	✓	-	-	✓	Continuous + codec	Qwen Thinker + speech-token Talker + vocoder	2025.03
LLaMA-Omni 2	System	✓	-	-	-	✓	-	-	-	✓	Discrete units	Audio encoder + LLaMA + streaming unit decoder	2025.05
VoxMind	System	△	-	-	-	✓	-	-	-	✓	Speech tokens	Speech tokenizer + agentic dialogue LM + speech decoder	2026.07
ZipVoice-Dialog	Model	△	-	-	-	✓	-	-	-	✓	Continuous latent	Flow-matching DiT speech generator	2026.07
Interaction Mode: Half-duplex / Controlled Barge-in													
Incremental TT Model	Component	✓	-	-	-	-	-	-	-	-	Incremental features	Incremental neural TT predictor (no generator)	2019.09
Listener-aware BC	Component	△	-	-	✓	-	-	-	-	-	ASR text	Listener-aware neural BC predictor (no generator)	2020.05
BPM_MT	Component	△	-	-	✓	-	-	-	-	-	Speech + text	Multitask neural BC predictor (no generator)	2021.11
Device Directedness	Component	△	-	-	-	-	-	-	-	-	Acoustic + context	Acoustic-context directedness classifier	2022.11
Acoustic Barge-in	Component	△	✓	-	-	-	-	-	-	-	Acoustic + context	Contextual acoustic barge-in classifier	2022.09
Freeze-Omni	System	✓	✓	△	△	✓	-	-	-	✓	Continuous + units	Speech encoder + frozen LLM + chunkwise unit decoder	2024.11
Mini-Omni2	System	✓	✓	△	△	✓	✓	-	-	✓	SNAC codec	Whisper encoder + Qwen + SNAC token decoder	2024.10
VITA	System	✓	✓	△	△	✓	✓	-	✓	-	Encoder + TTS	Audio encoder + LLM + streaming TTS	2024.08
SHANKS	Add-on	✓	△	△	-	✓	-	-	✓	-	Streaming features	Streaming speech encoder + side-path state predictor	2025.10
Interaction Mode: Full-duplex													
Duplex Conversation (Outbound)	System	✓	✓	✓	△	✓	-	✓	-	-	ASR / TTS	Streaming ASR + dialogue controller + TTS	2021.08
Duplex Conversation	System	✓	✓	✓	✓	✓	-	-	✓	-	ASR / TTS	Two-channel interaction model + ASR/TTS	2022.05
Synchronous LLM	Model	✓	✓	✓	✓	✓	-	-	-	✓	Speech units	Dual-channel speech units + synchronous LLaMA	2024.09
Moshi	Model	✓	✓	✓	✓	✓	-	-	-	✓	Mimi codec	Mimi codec + Helium temporal/depth Transformers	2024.10
SALMONN-omni	Model	✓	✓	✓	✓	✓	-	-	-	✓	Continuous embeddings	Speech encoder + codec-free asynchronous speech decoder	2024.11
Full-duplex LLM Scheme	System	✓	✓	✓	△	✓	-	✓	-	-	ASR / LLM / TTS	Streaming ASR + LLM dialogue manager + TTS	2024.05
MinMo	Model	✓	✓	✓	✓	✓	✓	-	-	✓	Speech tokens	Speech encoder/tokenizer + 8B LLM + speech decoder	2025.01
SALM-Duplex	Model	✓	✓	✓	△	✓	-	-	-	✓	Encoder + codec	Streaming speech encoder + LLM + codec decoder	2025.05
OmniFlatten	Model	✓	✓	✓	✓	✓	-	-	-	✓	Speech-text tokens	Flattened speech-text streams + decoder-only Transformer	2025.07
Japanese FD System	System	✓	✓	✓	△	✓	-	✓	-	-	ASR / LLM / TTS	Streaming ASR + LLM dialogue manager + TTS	2025.08
FireRedChat	System	✓	✓	✓	△	✓	-	✓	✓	-	ASR / TTS or codec	ASR/TTS cascade or codec-based semi-cascade	2025.09
FD Dialogue + DST	System	✓	✓	✓	△	✓	-	✓	✓	-	ASR / DST / TTS	Streaming ASR + DST/dialogue manager + TTS	2025.12
Chronological Thinking	Model	✓	✓	✓	✓	✓	-	-	-	✓	Synchronized tokens	Synchronized speech tokens + chronological Transformer	2025.10
LLM-Enhanced FD-DM	System	✓	✓	✓	△	✓	-	✓	-	-	ASR / LLM-DM / TTS	Streaming ASR + LLM-enhanced dialogue manager + TTS	2025.02
LSLM	Model	✓	✓	✓	△	✓	-	-	-	✓	SSL + speech tokens	SSL speech encoder + channel-fusion LLM + token decoder	2025.02
Chroma 1.0	System	✓	✓	✓	△	✓	-	-	-	✓	Speech tokens	Neural codec + real-time dialogue Transformer	2026.01
PersonaPlex	Model	✓	✓	✓	△	✓	-	-	-	✓	Neural codec	Mimi codec + duplex Transformer + voice conditioning	2026.02
MiniCPM-o 4.5	Model	✓	✓	✓	△	✓	✓	-	-	✓	Omni-modal stream	Omni encoder + MiniCPM LLM + streaming speech decoder	2026.04
DuplexSLA	Model	✓	✓	✓	✓	✓	-	-	-	✓	Continuous + discrete	Speech encoder + joint language/action LM + codec decoder	2026.05
BayLing-Duplex	Model	✓	✓	✓	✓	✓	-	-	-	✓	Audio tokens	Audio tokenizer + autoregressive duplex LLM + decoder	2026.06
F-Actor	Add-on	✓	✓	✓	✓	✓	-	-	✓	-	Control states	Behavior-state controller (uses host speech model)	2026.07
Hierarchical A-S Modeling	Model	✓	✓	✓	✓	✓	-	-	-	✓	Semantic + acoustic	Hierarchical semantic/acoustic duplex Transformer	2026.07
JoyAI-Talker	Model	✓	✓	✓	△	✓	-	-	-	✓	Semantic + acoustic	Semantic/acoustic tokenizers + dual-level Transformer	2026.08
SoulX-Duplug	Add-on	✓	✓	✓	△	✓	-	-	✓	-	Streaming ASR	Streaming ASR + semantic-state predictor (host TTS)	2026.03

Abbreviations: ASR, automatic speech recognition; TTS, text-to-speech; SSL, self-supervised learning; LLM, large language model; VAD, voice activity detection. Acoustic / Speech Model reports the principal speech encoder, tokenizer, dialogue backbone, decoder, or controller evaluated by each paper; Release uses YYYY.MM and follows the cited archival publication month, or the first arXiv month when the cited work is a preprint.

Development Trajectory: From Spoken Dialogue Models to Voice Agents

How to read the trajectory. Duplex task systems provide engineering context, while dGSLM (March 2022) marks the opening of the generative spoken-dialogue-model track. Work-type tags separate complete models/systems from foundations, add-ons, methods, and benchmarks; the three analytical columns then trace architecture/representation, interaction taxonomy, and agency/evaluation without attributing every stage-level claim to every listed work.

Phase	Period	Representative works	Architecture / Representation	Interaction Taxonomy	Agency / Evaluation Frontier
Engineering precursor	2021.08	[System] Duplex Conversation (Outbound)	Cascaded ASR-dialogue-policy-TTS stack with an explicit interruption and control layer.	Controller-mediated duplex / barge-in; listening and speaking states are externally orchestrated.	Domain-bound task completion; no unified generative spoken-dialogue model.
Generative model-track opening	2022.03-05	[Model] dGSLM; [System] Duplex Conversation	Textless dual-channel speech generation and learned conversational timing complement a controller-based duplex system.	Turn-taking plus learned overlap and backchannel timing; duplex becomes an explicit modeling target.	Open-domain spoken generation emerges; instruction following and tool use remain absent.
Speech-language foundations	2023	[Model] SpeechGPT; [Foundation] AudioPaLM	Discrete speech units, speech-text alignment, and cross-modal language-model adaptation.	Predominantly turn-based speech input/output; AudioPaLM is supporting speech-language work, not a dialogue-specific system.	Spoken instruction following improves, while real-time synchrony and multi-turn agency remain limited.
Real-time and duplex diversification	2024.05-11	[System] Full-duplex LLM Scheme; [System] VITA; [System] Mini-Omni; [Model] Synchronous LLMs; [Model] Moshi; [System] Mini-Omni2; [Model] SALMONN-omni	Cascaded real-time schemes diversify into streaming/interleaved generation and synchronous dual-stream speech models.	Controlled interruption expands toward native full-duplex overlap, backchannels, and simultaneous listening/speaking.	Conversation becomes more fluid; external action and persistent planning are not yet central.
Scaled unified voice models	2025.01-09	[Model] MinMo; [System] Step-Audio; [System] Baichuan-Audio; [Model] Qwen2.5-Omni; [Model] VITA-Audio; [Model] SALM-Duplex; [Model] OmniFlatten; [System] FireRedChat	Unified omni-modal LLMs, interleaved audio/text tokens, plus end-to-end, cascaded, and semi-cascaded variants.	Low-latency streaming scales alongside increasingly robust full-duplex models such as MinMo and OmniFlatten.	Instruction following and robustness scale up; tool orchestration is still mostly external.
Tool augmentation and agentic evaluation	2025.10	[Method] Stream RAG; [Benchmark] VoiceAgentBench	Stream RAG adds predictive streaming tool use; VoiceAgentBench supplies agentic evaluation rather than a model architecture.	Tool calls can begin during incoming speech, while evaluation covers multi-turn and multi-tool spoken workflows.	Tool selection, parameter filling, workflow completion, robustness, and safety become explicit targets.
Full-duplex control and evaluation	2026.02-04	[Model] PersonaPlex; [Add-on] SoulX-Duplug; [Model] MiniCPM-o 4.5; [Benchmark] Full-Duplex-Bench-v3	Persona conditioning and omni-modal generation are paired with a host-dependent state-control add-on and benchmark evidence.	Dynamic turn state, interruption, overlap, role control, and disfluency-aware evaluation are jointly emphasized.	The focus shifts from fluent duplex behavior toward controllability and reliable tool use under realistic speech.
Reasoning and action integration	2026.05-07	[Model] DuplexSLA; [System] VoxMind; [Method] Dual-Reasoner	DuplexSLA synchronizes speech/language/action; VoxMind adds Think-before-Speak planning and tool management; Dual-Reasoner targets thinking-while-talking.	The stage spans synchronized speech-action, sequential pre-speech planning, and interleaved reasoning during streaming speech.	Reason-plan-act behavior is integrated with speech; memory, safety, and long-horizon autonomy remain open.

Trajectory Synthesis

- Representation: cascaded ASR/TTS -> textless dual-channel units -> speech-native LLMs -> parallel semantic/acoustic streams -> synchronized speech-language-action generation.
- Interaction: controller-mediated barge-in -> learned turn timing -> streaming interruption -> full-duplex overlap/backchannels -> synchronized or interleaved speech, reasoning, and action.
- Agency: domain task logic -> generative dialogue -> spoken instruction following -> retrieval/tool use -> reason-plan-act behavior and persona control.
- Evidence: speech quality and timing -> latency/interruption -> overlap naturalness -> tool correctness and task completion -> memory, safety, and agent reliability.

Type key. [Model/System] complete dialogue implementation; [Foundation] supporting speech-language work not specific to multi-turn dialogue; [Add-on] host-dependent module; [Method] training or tool-use method; [Benchmark] evaluation evidence.

Development Roadmap and Capability Curve

The main curve summarizes a qualitative progression from generative spoken-dialogue modeling to integrated spoken agents. Year branches expand representative works from this collection: color denotes the primary interaction or agency path rather than an exhaustive capability set, while dashed tags identify benchmark or training evidence.

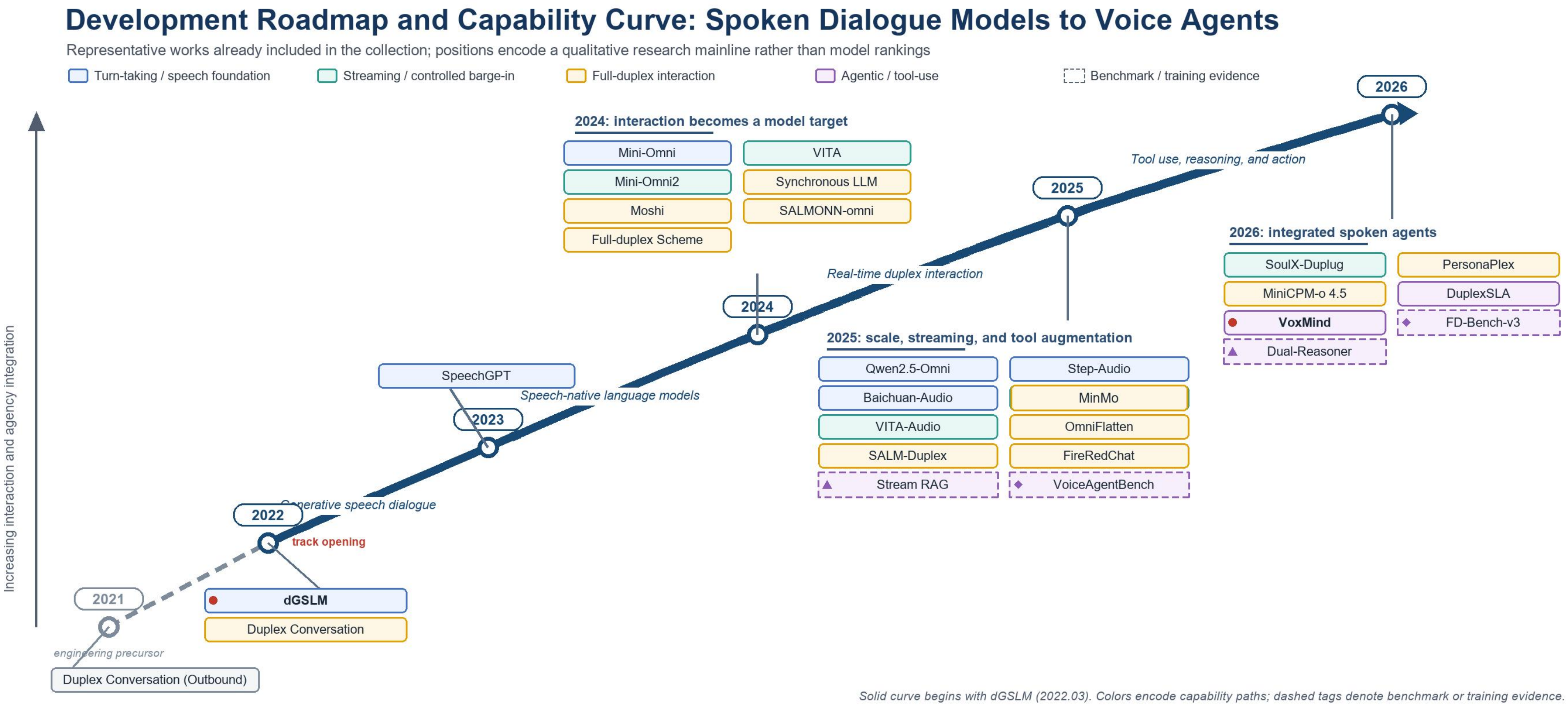

Figure 1. Academic roadmap and conceptual capability curve for Multi-Turn Speech Dialogue. The dotted 2021 segment provides engineering context; the solid model-development mainline begins with dGSLM in March 2022. Colors encode each work's primary contribution path, and dashed tags distinguish benchmark or training evidence from complete systems.

Interpretation. The field evolves along three coupled axes: speech-native representation, increasingly synchronous interaction, and growing task agency. The dense expansion after 2024 reflects diversification rather than a single winning architecture; by 2025-2026, retrieval, tool use, reasoning, action generation, and agent-oriented evaluation begin to close the loop from model-centric conversation to task-capable voice agents.